DEFEROXAMINE

Iron overdose & transfusion overload

Rapid IV → hypotension

CHEMICAL NAME N′-[5-(acetylhydroxyamino)pentyl]-N-[5-[[4-[(5-aminopentyl)hydroxyamino]-1,4-dioxobutyl]amino]pentyl]-N-hydroxybutanediamide (mesilate) — desferrioxamine

CLASS Iron-chelating agent (antidote)

TRADE NAMES Desferal · Desferrioxamine mesilate · Deferoxamine injection

THERAPEUTIC USES / INDICATIONS

- Acute iron poisoning (severe toxicity, serum iron >90 µmol/L (500 mcg/dL), shock, acidosis).
- Chronic transfusional iron overload (thalassaemia major, sickle cell, MDS).
- Aluminium overload in dialysis patients.

ADMINISTRATION

- **IV** infusion (acute poisoning) — rate-limited; **SC** infusion over 8–12 h via portable pump, 5–7 nights/week (chronic overload).
- IM possible (less effective).
- Rapid IV → flushing, hypotension, shock.

DOSAGES

- **Acute iron poisoning:** 15 mg/kg/h IV, reduce after 4–6 h; max 80 mg/kg/24 h (usually ≤24 h).
- **Chronic overload:** 20–60 mg/kg/day SC over 8–12 h, 5–7 days/week (with vitamin C 2–3 mg/kg after starting infusion in some protocols).

PHARMACOKINETICS

- Poor oral absorption — parenteral only.
- Ferrioxamine (iron complex) excreted in urine (reddish-orange "vin rosé" urine) and bile. **t½:** 6 h.

PHARMACODYNAMICS

- Binds **ferric iron (Fe^{3+})** from ferritin and haemosiderin (and free iron) → forms stable water-soluble ferrioxamine → excreted.
- Also binds aluminium.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Reddish-orange (vin rosé) urine.
- Injection-site pain, swelling (SC).
- Flushing, urticaria.

ADVERSE EFFECTS (SERIOUS)

- Hypotension, shock (rapid IV).
- Ototoxicity and retinal toxicity (long-term, high dose relative to iron load).
- Yersinia and mucormycosis infections (iron siderophore for organisms).
- ARDS (prolonged high-dose IV >24 h); growth retardation in children; renal impairment.

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity.
- **Caution:** severe renal impairment (unless dialysed), pregnancy.

KEY INTERACTIONS

- **Vitamin C (high dose):** ↑ iron toxicity to heart if started before chelation.
- **Prochlorperazine:** loss of consciousness (reported).

NURSING MANAGEMENT

- Acute poisoning: monitor BP, urine colour/output, serum iron, acid–base; supportive care.
- Chronic: teach SC pump use; audiometry and eye examination annually.
- Monitor ferritin, growth (children), renal function.
- Report fever, diarrhoea, abdominal pain (Yersinia).

PATIENT EDUCATION

- Urine may turn reddish — this shows iron is being removed.
- Report hearing or vision changes, fever, abdominal pain.
- Keep regular hearing and eye checks.

KEY POINTS

- Iron chelator; antidote and transfusion overload.
- Vin rosé urine.
- Acute: 15 mg/kg/h IV (not faster).
- Oto/retinal toxicity; Yersinia infection.